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C C E C C P R O C E E D I N G S

**Advancing Clean Energy
Solutions for Climate-
Resilient and Sustainable
Development in Nigeria**

Rethinking clean energy utilisation for climate change mitigation and sustainable development in Nigeria: The imperative of community inclusion, mass education and public policy initiatives

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Abstract

The accelerating impacts of climate change, driven largely by dependence on fossil fuels, have heightened the urgency for a transition to clean energy systems, particularly in developing countries like Nigeria. Despite Nigeria's vast renewable energy potential especially in solar, wind, and biomass clean energy utilisation remains significantly low due to structural, socio-economic, and policy-related constraints. This study rethinks clean energy utilisation as a critical pathway for climate change mitigation and sustainable development, emphasizing the imperative of community inclusion, mass education, and coherent public policy initiatives. Adopting a qualitative and analytical approach, the paper draws on secondary data sources, including policy documents, scholarly literature, and reports from international development agencies. The study interrogates the existing gaps in clean energy adoption in Nigeria, identifying key barriers such as limited public awareness, weak institutional frameworks, inadequate financing mechanisms, and minimal community participation in energy planning and implementation. It argues that top-down energy policies have largely failed to achieve widespread adoption due to insufficient grassroots engagement and lack of context-specific education strategies. Findings reveal that communities with higher levels of awareness and participatory involvement demonstrate greater acceptance and utilisation of clean energy technologies, thereby contributing to reduced carbon emissions and improved livelihoods. The study concludes that aligning clean energy strategies with community-driven approaches and sustained educational campaigns can significantly enhance Nigeria's capacity to mitigate climate change while achieving long-term sustainability goals. It recommends the institutionalization of community-based energy programs, integration of clean energy education into formal and informal systems, and the strengthening of policy coherence to support an inclusive energy transition.

Keywords

Clean energy, climate change mitigation, sustainability, community inclusion, mass education, public policy, Nigeria.

1. Introduction

Climate change has emerged as one of the most pressing global challenges of the 21st century, with far-reaching implications for environmental sustainability, economic stability, and human well-being. The increasing concentration of greenhouse gases (GHGs), largely driven by the combustion of fossil fuels such as coal, oil, and natural gas, has led to rising global temperatures, extreme weather events, and ecological degradation (Intergovernmental Panel on Climate Change [IPCC], 2021). For developing countries like Nigeria, the consequences of climate change are particularly severe, manifesting in desertification, flooding, erratic rainfall patterns, declining agricultural productivity, and heightened energy insecurity (Adelekan, 2016; Federal Ministry of Environment, 2020). These challenges underscore the urgent need for sustainable and climate-resilient energy alternatives.

Clean energy utilisation has gained global recognition as a viable pathway for mitigating climate change while promoting sustainable development. Clean energy, often derived from renewable sources such as solar, wind, hydro, and biomass, offers the dual advantage of reducing carbon emissions and enhancing energy access (International Energy Agency [IEA], 2022). Nigeria is endowed with abundant renewable energy resources, particularly solar radiation, given its geographical location within the tropical belt (Ohunakin et al., 2014). Despite this potential, the country's energy mix remains heavily dominated by fossil fuels, accounting for a significant share of electricity generation and contributing to environmental degradation (Energy Commission of Nigeria, 2019).

The paradox of energy poverty amidst resource abundance persists in Nigeria, where a substantial proportion of the population lacks access to reliable and affordable electricity (World Bank, 2021). Rural communities are disproportionately affected, often relying on traditional biomass such as firewood and charcoal for cooking and heating, which exacerbates deforestation, indoor air pollution, and health risks (UNDP, 2020). This situation highlights the critical need to rethink existing approaches to energy utilisation by prioritizing clean, accessible, and community-driven solutions.

While policy frameworks and institutional efforts have been introduced to promote renewable energy adoption in Nigeria, progress has been relatively slow and uneven. Initiatives such as the National Renewable Energy and Energy Efficiency Policy (NREEEP) and various rural electrification programs have not fully achieved their intended outcomes due to implementation challenges, weak governance structures, and limited stakeholder engagement (Sambo, 2009; Aliyu et al., 2015). A major limitation of these initiatives is their predominantly top-down orientation, which often neglects the socio-cultural and economic realities of local communities.

Emerging literature suggests that sustainable energy transitions are more effective when local communities are actively involved in the planning, implementation, and management of energy projects (Sovacool, 2014). Community inclusion fosters a sense of ownership, enhances acceptance of new technologies, and ensures that interventions are tailored to local needs and contexts. In the Nigerian context, however, community participation in energy decision-making remains limited, thereby constraining the scalability and sustainability of clean energy initiatives.

In addition to community inclusion, mass education plays a pivotal role in driving clean energy adoption. Public awareness and energy literacy are essential for influencing attitudes, behaviors, and consumption patterns related to energy use (Ockwell et al., 2009). In many parts of Nigeria, low levels of awareness about the benefits, availability, and usability of clean energy technologies hinder their widespread acceptance. Educational interventions both formal and informal can serve as catalysts for behavioral change, empowering individuals and communities to make informed energy choices.

Furthermore, the role of public policy in shaping energy transitions cannot be overstated. Effective policy frameworks provide the necessary regulatory environment, financial incentives, and institutional support required to accelerate clean energy deployment (IEA, 2022). In Nigeria, policy inconsistencies, inadequate funding, and weak enforcement mechanisms have limited the effectiveness of existing energy policies. There is, therefore, a need for more coherent, inclusive, and adaptive policy approaches that integrate community perspectives and prioritize long-term sustainability.

Nigeria with vast renewable energy potential, yet the adoption and utilization of clean energy technologies remain significantly low, this has hindered progress toward climate change mitigation and sustainable development goals. The low utilization of clean and renewable energy can be attributed to many factors. These factors include but not limited to widespread deficiency in public knowledge and understanding of clean energy benefits, resulting in poor acceptance, limited behavioral change, and weak grassroots support for energy transition initiatives. Equally, clean energy projects in Nigeria are often designed and implemented without meaningful involvement of local communities, leading to poor project sustainability, resistance, and lack of local ownership. Closely related, is the existing energy policies and regulatory frameworks that are often fragmented, inconsistently implemented, and lack enforcement mechanisms, thereby undermining effective clean energy development and climate action. This can be attributed to bureaucratic inefficiencies which has made Nigerian's to rely on fossil fuels thereby, continued to exacerbate the degradation impact of the environment. In addition, the high cost of renewable energy technologies, coupled with inadequate infrastructure and limited access to financing, restricts large-scale deployment and accessibility, especially in most communities in the rural areas and even the urban slums. The gap between urban and rural areas continue to widen in accessing clean, affordable and reliable energy.

It is against this backdrop, this study seeks to rethink clean energy utilisation in Nigeria as a strategic instrument for climate change mitigation and sustainable development. It emphasizes the interconnected roles of community inclusion, mass education, and public policy initiatives in driving an inclusive and effective energy transition. By addressing existing gaps in awareness, participation, and policy implementation, the study contributes to ongoing discourse on sustainable energy development and offers practical insights for enhancing clean energy adoption in Nigeria.

Ultimately, achieving a sustainable energy future in Nigeria requires a paradigm shift from conventional, centralized energy models to more inclusive, participatory, and knowledge-driven approaches. Such a transition not only holds the potential to mitigate climate change impacts but also to improve livelihoods, reduce poverty, and foster national development.

The broad objective of this study is to examine clean energy utilisation as a strategic instrument for climate change mitigation and sustainable development in Nigeria, with particular emphasis on the roles of community inclusion, mass education, and public policy initiatives in promoting an inclusive and sustainable energy transition.

The study specifically seeks to:

1. examine the role of clean energy utilisation in promoting climate change mitigation and sustainable development in Nigeria.
2. assess the influence of community inclusion and mass education on the adoption, acceptance, and sustainability of clean energy initiatives in Nigeria.
3. evaluate the effectiveness of public policy initiatives and institutional frameworks in enhancing clean energy utilisation and facilitating sustainable energy transition in Nigeria.

This study is significant because it contributes to the growing discourse on clean energy utilisation, climate change mitigation, and sustainable development in Nigeria by providing a holistic perspective that integrates community inclusion, mass education, and public policy initiatives. The study expands existing knowledge by demonstrating that effective clean energy transition is not solely dependent on technological advancement, but also on public awareness, participatory governance, and institutional effectiveness. It further enriches academic literature within the fields of public administration, environmental governance, sustainable development, and energy policy by establishing the interconnected relationship between clean energy adoption, social participation, and sustainability outcomes.

The study is also significant to policymakers, development agencies, energy stakeholders, and community-based organizations as it provides practical insights for improving clean energy adoption and addressing energy poverty in Nigeria. The findings and recommendations of the study will assist government institutions in designing more inclusive and effective renewable energy policies, while guiding development practitioners on the importance of grassroots participation and educational interventions in energy projects.

2. Conceptual Clarification

This section provides detailed clarification of the key concepts underpinning this study. These concepts are essential for understanding the relationships between clean energy utilisation, climate change mitigation, sustainability, community inclusion, mass education, and public policy within the Nigerian context.

2.1 Clean Energy

Clean energy refers to energy derived from sources that produce little or no environmental pollution or greenhouse gas emissions. It is closely associated with renewable energy sources such as solar, wind, hydro, geothermal, and biomass. Unlike fossil fuels, clean energy systems are environmentally sustainable and contribute minimally to global warming (International Energy Agency, 2022). From a conceptual standpoint, clean energy is not only defined by its

source but also by its environmental impact and sustainability over time. According to the International Energy Agency (2022), clean energy plays a critical role in reducing carbon emissions and facilitating the transition toward low-carbon economies. In Nigeria, clean energy utilisation primarily involves solar photovoltaic systems, mini-grids, and biomass alternatives aimed at improving energy access while minimizing ecological damage (Ohunakin et al., 2014). In the context of this study, clean energy is conceived as a source of energy that is affordable and accessible to the teeming population which is sustainable via a period of time through deliberate public policy initiatives.

2.2 Climate Change Mitigation

Climate change mitigation refers to efforts aimed at reducing or preventing the emission of greenhouse gases into the atmosphere. It involves strategies such as transitioning to renewable energy, improving energy efficiency, and adopting sustainable land-use practices (IPCC, 2021). The Intergovernmental Panel on Climate Change defines mitigation as human interventions that reduce sources or enhance the sinks of greenhouse gases (IPCC, 2021). Clean energy utilisation is a central mitigation strategy because it replaces carbon-intensive energy sources with environmentally friendly alternatives. In Nigeria, mitigation efforts are particularly important given the country's vulnerability to climate impacts such as flooding, desertification, and rising temperatures (Adelekan, 2016).

2.3 Sustainability

Sustainability refers to the ability to meet present needs without compromising the capacity of future generations to meet their own needs. This concept gained global recognition through the World Commission on Environment and Development (1987). Sustainability is typically understood through three interconnected pillars: environmental sustainability (protection of natural resources), economic sustainability (long-term economic growth), and social sustainability (equity and improved quality of life) (Sachs, 2015). In the context of this study, clean energy utilisation contributes to sustainability by reducing environmental degradation, promoting economic opportunities, and improving social welfare through enhanced energy access.

2.4 Clean Energy Utilisation

Clean energy utilisation refers to the adoption, deployment, and effective use of renewable and low-carbon energy technologies in meeting energy needs. It goes beyond availability to include accessibility, affordability, and efficiency in energy consumption (IEA, 2022). In Nigeria, clean energy utilisation includes the use of solar home systems, mini-grid electrification, wind energy systems, and improved biomass technologies. However, utilisation remains low due to financial, infrastructural, and socio-cultural barriers (Aliyu et al., 2015). Conceptually, it involves both technical adoption and behavioral acceptance, making it a socio-technical process rather than purely a technological one (Sovacool, 2014).

2.5 Community Inclusion

Community inclusion refers to the active involvement of local populations in decision-making processes related to development initiatives. It emphasizes participation, empowerment, and ownership of projects by beneficiaries (Chambers, 1997).

Within the framework of development studies, community inclusion is closely linked to participatory approaches, which argue that local knowledge and stakeholder engagement are critical for the success of development interventions. In clean energy projects, community inclusion ensures that technologies are tailored to local needs, thereby enhancing acceptance, sustainability, and long-term impact (Bhattacharyya, 2012).

In Nigeria, limited community inclusion has been identified as a key factor contributing to the failure of many energy projects (Akinwale & Ogundari, 2017). Therefore, integrating local stakeholders into planning and implementation processes is essential for improving clean energy adoption.

2.6 Mass Education

Mass education refers to the widespread dissemination of knowledge and information to a large population through formal and informal means. It includes structured education systems, media campaigns, public awareness programs, and community-based learning initiatives.

In the context of clean energy utilisation, mass education serves as a tool for increasing awareness of renewable energy technologies, promoting energy-efficient behaviors, and reducing misconceptions and resistance to innovation (Ockwell et al., 2009).

Drawing from Diffusion of Innovations (Rogers, 2003), mass education facilitates the spread of new technologies by enhancing knowledge and influencing attitudes. In Nigeria, low levels of energy literacy highlight the need for targeted educational interventions to accelerate clean energy adoption (Akinbami et al., 2013).

2.7 Public Policy

Public policy refers to the principles, regulations, and actions adopted by governments to address societal issues and guide development. It encompasses laws, programs, and institutional frameworks that shape economic and social activities (Dye, 2013).

In the energy sector, public policy plays a critical role in regulating energy production and distribution, providing incentives for renewable energy investments, and ensuring equitable access to energy resources (IEA, 2022).

In Nigeria, policies such as the National Renewable Energy and Energy Efficiency Policy aim to promote clean energy development. However, conceptual analysis shows that the effectiveness of such policies depends largely on implementation, institutional capacity, and stakeholder engagement (Energy Commission of Nigeria, 2019).

2.8 Energy Poverty

Energy poverty refers to the lack of access to reliable, affordable, and sustainable energy services. It is a multidimensional concept that affects households' ability to meet basic needs such as lighting, cooking, heating, and communication (World Bank, 2021).

According to the World Bank (2021), energy poverty is a major constraint to development, particularly in sub-Saharan Africa. In Nigeria, millions of people especially in rural areas lack access to electricity and rely on traditional biomass, which has negative environmental and health implications (UNDP, 2020).

Clean energy utilisation is widely regarded as a viable solution to energy poverty, as decentralized renewable systems can provide affordable and accessible energy to underserved communities.

2.9 Sustainable Development

Sustainable development is a holistic concept that integrates environmental protection, economic growth, and social inclusion. It is operationalized globally through initiatives such as the United Nations Sustainable Development Goals (SDGs), particularly Goal 7 (Affordable and Clean Energy) and Goal 13 (Climate Action) (UN, 2015).

In this study, sustainable development serves as the overarching goal that connects all other concepts. Clean energy utilisation contributes directly to achieving sustainable development by addressing energy needs while mitigating climate change and promoting inclusive growth.

3 Literature Review

3.1 Clean Energy Utilisation and Climate Change Mitigation

The linkage between clean energy utilisation and climate change mitigation is well established in the literature. According to the Intergovernmental Panel on Climate Change (IPCC, 2021), transitioning from fossil fuel-based energy systems to renewable energy sources is central to reducing greenhouse gas emissions and limiting global warming. Clean energy technologies such as solar, wind, and hydropower produce little to no carbon emissions, making them essential tools in achieving global climate targets (International Energy Agency [IEA], 2022).

Scholars such as Sovacool (2014) argue that clean energy transitions not only address environmental concerns but also offer socio-economic benefits, including job creation, improved public health, and energy security. Similarly, Jacobson and Delucchi (2011) emphasize that a global shift to renewable energy systems is technically feasible and economically viable, provided there is strong political will and societal support. In the African context, Onyeji et al. (2012) note that renewable energy adoption can significantly reduce vulnerability to climate change while supporting sustainable development goals.

3.2 Renewable Energy Potential and Challenges in Nigeria

Nigeria is richly endowed with renewable energy resources, yet its utilisation remains suboptimal. Studies by Ohunakin et al. (2014) and Sambo (2009) highlight the country's vast solar energy potential, with average solar radiation levels ranging between 3.5–7.0 kWh/m²/day. Wind energy prospects have also been identified in northern regions, while hydropower and biomass resources are abundant across the country (Energy Commission of Nigeria, 2019).

Despite this potential, several barriers hinder effective clean energy utilisation. Aliyu et al. (2015) identify key challenges, including inadequate infrastructure, high initial investment costs, limited technical capacity, and weak institutional frameworks. Furthermore, energy poverty remains a critical issue, with millions of Nigerians lacking access to reliable electricity (World Bank, 2021). Rural communities, in particular, rely heavily on traditional biomass, which contributes to environmental degradation and health hazards (UNDP, 2020).

Adelekan (2016) further notes that climate change exacerbates existing vulnerabilities in Nigeria, especially in agriculture-dependent communities. The slow pace of renewable energy adoption, therefore, limits the country's ability to mitigate these impacts effectively.

3.3 Community Inclusion in Clean Energy Transition

The importance of community inclusion in sustainable energy development has been widely emphasized in contemporary literature. Participatory approaches to energy planning are considered more effective in ensuring the sustainability and acceptance of clean energy initiatives (Sovacool, 2014). When local communities are actively involved in decision-making processes, they are more likely to adopt and maintain renewable energy technologies. Walker and Devine-Wright (2008) introduced the concept of “energy justice,” which underscores the need for equitable distribution of energy resources and inclusive participation in energy governance. In the Nigerian context, Akinwale and Ogundari (2017) argue that the exclusion of local communities from energy projects often leads to resistance, low adoption rates, and project failure.

Empirical studies also demonstrate that community-based energy projects tend to be more successful due to increased trust, local knowledge integration, and shared ownership (Bhattacharyya, 2012). However, in Nigeria, energy policies and projects are often centrally designed, with minimal input from grassroots stakeholders, thereby limiting their effectiveness and sustainability.

3.4 Mass Education and Energy Literacy

Mass education plays a crucial role in promoting clean energy adoption by enhancing public awareness and influencing behavioral change. Ockwell et al. (2009) assert that knowledge dissemination and capacity building are essential components of successful energy transitions. Without adequate understanding of clean energy technologies, their benefits, and usage, communities are less likely to embrace them.

In Nigeria, low levels of energy literacy remain a significant barrier to renewable energy adoption (Akinbami et al., 2013). Many individuals are unaware of the long-term cost savings and environmental benefits associated with clean energy solutions. Educational interventions ranging from formal curricula integration to community outreach programs can help bridge this knowledge gap.

Furthermore, Rogers' (2003) Diffusion of Innovations theory highlights the importance of communication channels and social systems in the adoption of new technologies. Applying this framework, mass education can accelerate the diffusion of clean energy innovations by fostering awareness, interest, and eventual acceptance among diverse population groups.

3.5 Public Policy and Institutional Frameworks

Public policy is a critical driver of clean energy development, providing the regulatory and institutional support necessary for large-scale adoption. In Nigeria, several policy frameworks have been introduced to promote renewable energy, including the National Renewable Energy and Energy Efficiency Policy (NREEEP) and the Renewable Energy Master Plan (REMP) (Energy Commission of Nigeria, 2019).

However, the effectiveness of these policies has been limited by implementation challenges, policy inconsistencies, and inadequate funding (Aliyu et al., 2015). According to the International Energy Agency (IEA, 2022), successful clean energy transitions require coherent policy frameworks that integrate financial incentives, technological innovation, and stakeholder engagement.

Additionally, public-private partnerships (PPPs) have been identified as viable mechanisms for mobilizing resources and expertise in the energy sector (World Bank, 2021). Yet, in Nigeria, weak institutional capacity and governance issues continue to hinder the effectiveness of such collaborations.

Scholars also emphasize the need for inclusive policy design that incorporates local perspectives and addresses socio-economic realities (Sovacool, 2014). Policies that fail to consider the needs and constraints of end-users are less likely to achieve desired outcomes.

3.6 Research Gap

The reviewed literature underscores the critical role of clean energy utilisation in addressing climate change and promoting sustainable development. While Nigeria possesses significant renewable energy potential, multiple challenges ranging from infrastructural deficits to socio-cultural barriers continue to impede its effective utilisation.

Importantly, existing studies reveal a gap in the integration of community inclusion, mass education, and public policy as interconnected drivers of clean energy adoption. Much of the current approach remains fragmented, with limited emphasis on participatory frameworks and grassroots engagement. Therefore, this study seeks to bridge this gap by advancing a holistic perspective that combines community-driven approaches, educational interventions, and policy reforms as essential components of a sustainable energy transition in Nigeria.

4. Methodology

This study adopts a qualitative research design to examine clean energy utilisation as a strategy for climate change mitigation and sustainable development in Nigeria, with particular emphasis on community inclusion, mass education, and public policy initiatives. The methodology is structured to provide a comprehensive understanding of existing challenges, institutional dynamics, and pathways for enhancing clean energy adoption.

The study employs a descriptive and analytical research design, which allows for an in-depth exploration of the subject matter through the synthesis and interpretation of existing knowledge. This approach is appropriate given the conceptual and policy-oriented nature of the research, which seeks to rethink existing frameworks rather than generate primary empirical data. The design integrates elements of exploratory research, aimed at identifying gaps in current clean energy practices and proposing innovative solutions tailored to the Nigerian context.

The study relies exclusively on secondary data, drawn from a wide range of credible sources which includes academic journal articles and scholarly publications, government policy documents (e.g., National Renewable Energy and Energy Efficiency Policy), Reports from international organizations such as the World Bank, United Nations Development Programme (UNDP), and International Energy Agency (IEA), Publications from the Energy Commission of Nigeria & Federal Ministry of Environment and Conference papers and institutional reports relevant to clean energy and climate change. These sources provide both theoretical and empirical insights necessary for a robust analysis of clean energy utilisation in Nigeria.

Data were collected through systematic literature review and document analysis. Relevant materials were identified using key search terms such as clean energy, renewable energy in Nigeria, climate change mitigation, community participation, energy policy, and sustainable development. Priority was given to recent and peer-reviewed publications to ensure the reliability and relevance of the data.

Official policy documents and institutional reports were also critically examined to understand the existing regulatory and policy environment governing clean energy in Nigeria.

The study adopts a thematic content analysis approach to analyze the collected data. This involves:

Data familiarization: Careful reading and re-reading of selected materials

Coding and categorization: Identification of recurring themes such as energy access, policy gaps, community engagement, and education

Thematic synthesis: Grouping related ideas into coherent analytical categories

Interpretation: Drawing insights and establishing relationships between clean energy utilisation, climate change mitigation, and sustainability

This method enables the researcher to systematically interpret qualitative data and generate meaningful conclusions.

The study is anchored on an integrated analytical framework that combines:

- Sustainable Development Theory (emphasizing environmental, social, and economic balance)
- Participatory Development Approach (highlighting the role of community inclusion)
- Diffusion of Innovations Theory (Rogers, 2003), which explains how new technologies, such as renewable energy systems, are adopted within societies

This multi-theoretical approach provides a holistic lens for analyzing the dynamics of clean energy adoption in Nigeria.

The study focuses on Nigeria as a whole, with particular attention to rural and underserved communities where energy poverty is most prevalent. It examines national-level policies while also considering local-level realities that influence clean energy utilisation.

As a qualitative study based on secondary data, the research is subject to certain limitations such as; the study does not include field surveys or interviews, which may limit firsthand insights. Also, the differences in data quality and reporting standards across sources may affect consistency. Equally, findings may be influenced by the availability and scope of existing literature. However, despite these limitations, efforts were made to ensure the use of reliable and up-to-date sources to enhance the validity of the study.

This study is anchored on an integrated theoretical framework that draws from Sustainable Development Theory, Participatory Development Theory, and Diffusion of Innovations Theory. The combination of these theories provides a comprehensive lens for understanding clean energy utilisation as a tool for climate change mitigation and sustainability, particularly within the Nigerian context where socio-economic and institutional dynamics significantly influence energy transitions.

Sustainable Development Theory provides the foundational basis for this study, emphasizing the need to balance environmental protection, economic growth, and social equity. Popularized by the Brundtland Commission Report (1987), sustainable development is defined as development that meets present needs without compromising the ability of future generations to meet their own needs. In the context of clean energy utilisation, this theory underscores the importance of transitioning from fossil fuel-based energy systems to renewable and environmentally friendly alternatives. According to Sachs (2015), sustainable energy systems are central to achieving long-term development goals, particularly in developing economies where energy access remains limited. Clean energy contributes to environmental sustainability by reducing greenhouse gas emissions, supports economic development through job creation and innovation, and enhances social well-being by improving access to reliable energy services.

In all, the application of Sustainable Development Theory highlights the urgent need to address energy poverty while mitigating environmental degradation. The reliance on traditional biomass and fossil fuels not only contributes to climate change but also undermines public health and economic productivity (UNDP, 2020). Thus, clean energy utilisation aligns with the core principles of sustainability by promoting inclusive and environmentally responsible development.

Participatory Development Theory emphasizes the active involvement of local communities in the planning, implementation, and evaluation of development initiatives. The theory

challenges top-down approaches to development, advocating instead for bottom-up strategies that empower communities as key stakeholders (Chambers, 1997).

This theoretical perspective is particularly relevant to clean energy utilisation in Nigeria, where many energy projects have historically been designed and implemented without adequate community input. Such approaches often result in low adoption rates, resistance, and project failure (Akinwale & Ogundari, 2017). Participatory Development Theory argues that when communities are included in decision-making processes, they develop a sense of ownership and responsibility, which enhances the sustainability of interventions.

In relation to this study, community inclusion is viewed as a critical driver of clean energy adoption. By incorporating local knowledge, cultural values, and socio-economic realities, energy initiatives can be better tailored to meet the needs of end-users. Furthermore, participatory approaches facilitate trust-building and collaboration between policymakers, private sector actors, and local populations, thereby improving the effectiveness of clean energy programs.

The Diffusion of Innovations Theory, developed by Rogers (2003), explains how new ideas, technologies, or practices spread within a society over time. The theory identifies key elements of diffusion, including the innovation itself, communication channels, time, and the social system.

In the context of clean energy utilisation, renewable energy technologies such as solar panels and improved cookstoves can be considered innovations that require widespread acceptance and adoption. According to Rogers (2003), the rate of adoption is influenced by factors such as perceived benefits, compatibility with existing practices, complexity, trialability, and observability.

Mass education plays a crucial role in this process by increasing awareness and understanding of clean energy technologies. Ockwell et al. (2009) argue that information dissemination and capacity building are essential for overcoming resistance to new technologies. In Nigeria, low levels of energy literacy and limited access to information have slowed the diffusion of clean energy solutions (Akinbami et al., 2013).

By applying this theory, the study highlights the importance of communication strategies, education campaigns, and demonstration projects in accelerating the adoption of clean energy technologies. It also underscores the role of social networks and community leaders in influencing attitudes and behaviors toward energy use.

The integration of these three theories provides a holistic framework for analyzing clean energy utilisation in Nigeria: The Sustainable Development Theory establishes the overarching goal of achieving environmental, economic, and social balance through clean energy adoption. While the Participatory Development Theory emphasizes the need for inclusive, community-driven approaches to ensure the relevance and sustainability of energy initiatives and Diffusion of Innovations Theory explains the mechanisms through which clean energy technologies can be effectively adopted and scaled within society.

Together, these theories highlight that successful clean energy transitions are not solely dependent on technological availability but also on social inclusion, awareness creation, and

supportive policy environments. The interplay between these elements is critical for addressing the multifaceted challenges of climate change and energy poverty in Nigeria. This integrated theoretical framework is particularly suited to the objectives of this study, as it aligns with the central argument that clean energy utilisation in Nigeria must go beyond technical solutions to incorporate social, educational, and policy dimensions. It provides a structured basis for analyzing how community inclusion, mass education, and public policy initiatives can collectively drive sustainable energy transitions and climate change mitigation. By grounding the study in these theoretical perspectives, the research contributes to both academic discourse and practical policy formulation, offering insights into how Nigeria can achieve a more inclusive and sustainable energy future.

5. Findings

The findings of this study are derived from a comprehensive analysis of secondary data on clean energy utilisation, climate change mitigation, and sustainable development in Nigeria. Using a thematic content analysis approach, key insights are presented across major areas, including awareness and energy literacy, community participation, policy and institutional frameworks, infrastructural and financial constraints, and the overall implications for climate change mitigation and sustainability.

The study reveals that a significant barrier to clean energy utilisation in Nigeria is the low level of public awareness and energy literacy. Many individuals, particularly in rural communities, lack adequate knowledge of clean energy technologies such as solar home systems, mini-grids, and improved cookstoves. This knowledge gap contributes to resistance, misconceptions, and limited adoption of renewable energy solutions (Akinbami et al., 2013). Findings further indicate that where awareness campaigns and educational programs have been implemented, there is a noticeable increase in acceptance and utilisation of clean energy technologies. This underscores the critical role of mass education in shaping public attitudes and promoting behavioral change. However, existing educational efforts remain fragmented, uncoordinated, and insufficient in scale to drive nationwide transformation.

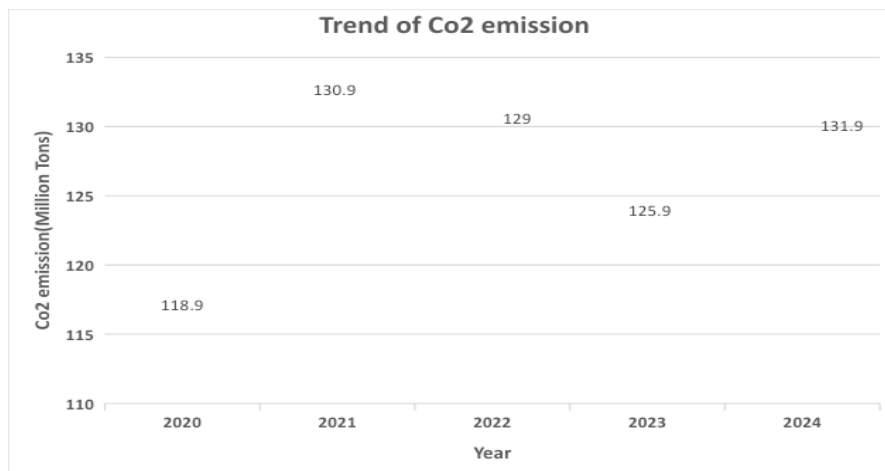


Figure 1: Trend of Co2 emissions in Nigeria (**Source:** Worldometer (2024), World Bank (2023))

Emissions fluctuate but remain consistently high, reflecting dependence on fossil fuels

Another major finding is the limited involvement of local communities in clean energy planning and implementation. Most energy projects in Nigeria are designed using a top-down approach, with minimal input from end-users. This often results in misalignment between project design and community needs, leading to low acceptance, poor maintenance, and eventual project failure (Akinwale & Ogundari, 2017).

The study finds that community-based and participatory energy initiatives tend to achieve better outcomes. In cases where communities are actively engaged, there is a stronger sense of ownership, improved sustainability of projects, and higher adoption rates. This highlights the importance of integrating participatory development approaches into energy policy and program design.

The analysis reveals that although Nigeria has developed several policy frameworks to promote renewable energy, their implementation has been largely ineffective. Policies such as the National Renewable Energy and Energy Efficiency Policy (NREEEP) suffer from inconsistencies, inadequate funding, and weak enforcement mechanisms (Energy Commission of Nigeria, 2019).

Institutional challenges, including bureaucratic inefficiencies, lack of coordination among agencies, and limited technical capacity, further hinder policy effectiveness. Additionally, the absence of strong regulatory incentives and clear implementation strategies discourages private sector investment in clean energy projects.

The high initial cost of clean energy technologies is identified as a major barrier to adoption, particularly among low-income households. While renewable energy systems offer long-term cost savings, the upfront investment required remains prohibitive for many Nigerians (Aliyu et al., 2015).

In addition, inadequate infrastructure such as limited grid connectivity, poor distribution networks, and lack of maintenance services constrains the scalability of clean energy solutions. The study also finds that access to financing options, such as credit facilities and subsidies, is limited, further restricting the ability of individuals and communities to invest in clean energy technologies.

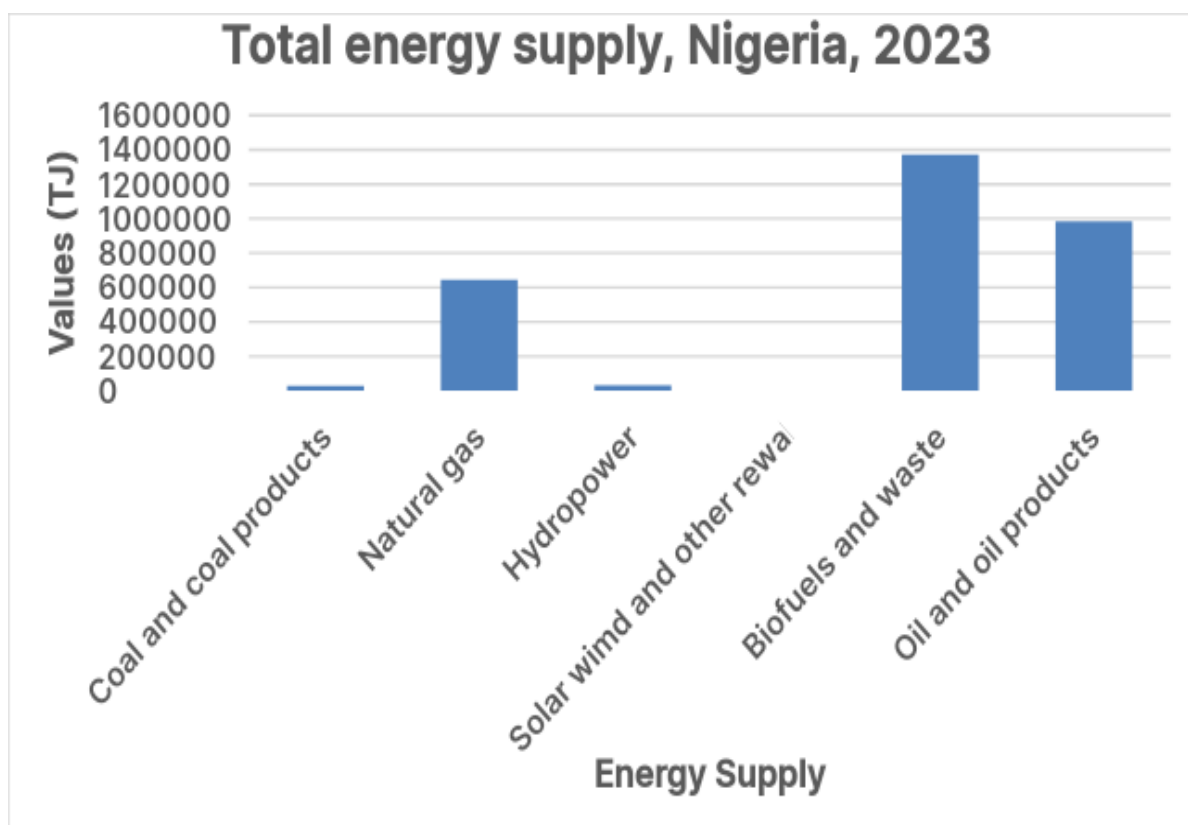


Figure 2: Total Energy Supply in Nigeria (**Source:** Worldometer (2024), World Bank (2023))

Establishes the problem of rising emission, forming the foundation for climate change mitigation discussion.

Despite Nigeria's renewable energy potential, the study finds that energy poverty remains widespread, particularly in rural areas. A large proportion of the population continues to rely on traditional biomass, such as firewood and charcoal, for cooking and heating (UNDP, 2020). This not only contributes to deforestation and environmental degradation but also poses serious health risks due to indoor air pollution.

The persistence of energy poverty highlights the failure of existing energy systems to meet the needs of the population and underscores the urgency of accelerating clean energy adoption.

Table 1: CO₂ emissions, per capital, and growth rate in Nigeria

Year	Co ₂ emission (Million Tons)	Co ₂ per Capital (Tons)	Growth Rate (%)
2020	118.9	0.58	- 5.7

2021	130.9	0.62	+10.08
2022	129.0	0.60	- 1.42
2023	125.9	0.57	- 2.46
2024	131.9	0.58	+ 4.80

Source: Worldometer (2024), World Bank (2023)

A key finding of this study is that integrated approaches combining community inclusion, mass education, and supportive public policies yield more effective outcomes in clean energy utilisation. Evidence from various programs indicates that when these elements are aligned, there is a significant improvement in technology adoption, user satisfaction, and environmental impact.

For instance, community-driven solar projects supported by awareness campaigns and policy incentives have demonstrated higher success rates compared to isolated interventions. This reinforces the argument that clean energy transitions must be approached holistically rather than through fragmented strategies.

The findings indicate that enhancing clean energy utilisation in Nigeria has significant potential to reduce greenhouse gas emissions and mitigate climate change impacts. Increased adoption of renewable energy can decrease dependence on fossil fuels and traditional biomass, thereby contributing to environmental sustainability.

Moreover, clean energy utilisation has broader socio-economic benefits, including improved livelihoods, job creation, and enhanced energy security. However, these benefits can only be fully realized if existing barriers are addressed through inclusive, educational, and policy-driven interventions.

5.1 Discussion of Findings

The findings of this study provide critical insights into the structural, social, and institutional dynamics shaping clean energy utilisation in Nigeria. This section interprets these findings in relation to existing literature and the study's theoretical framework, highlighting their implications for climate change mitigation and sustainable development.

The finding that low levels of awareness and energy literacy significantly hinder clean energy utilisation aligns with the Diffusion of Innovations Theory (Rogers, 2003), which emphasizes the role of knowledge and communication in the adoption of new technologies. The limited understanding of renewable energy technologies among Nigerians particularly in rural areas suggests that clean energy innovations have not effectively diffused across the social system. This supports the argument by Ockwell et al. (2009) that information dissemination and capacity building are essential for successful energy transitions. Without adequate knowledge of the benefits, cost implications, and operational mechanisms of clean energy technologies, potential users are less likely to adopt them. The observed increase in adoption

rates in areas with targeted awareness campaigns further reinforces the importance of mass education as a transformative tool.

From a practical standpoint, this implies that Nigeria's clean energy strategy must prioritize large-scale, coordinated educational initiatives. Integrating clean energy education into formal curricula and community outreach programs would not only enhance public understanding but also foster long-term behavioral change necessary for sustainable energy utilisation.

The study's finding on limited community participation highlights a critical weakness in Nigeria's current energy governance structure. This aligns with Participatory Development Theory (Chambers, 1997), which posits that development interventions are more effective and sustainable when beneficiaries are actively involved in decision-making processes.

The prevalence of top-down approaches in Nigeria's energy sector has resulted in projects that are often misaligned with local needs, leading to low acceptance and sustainability challenges. This corroborates the findings of Akinwale and Ogundari (2017), who argue that the exclusion of local communities undermines the success of development initiatives.

Conversely, the study shows that community-driven projects yield better outcomes due to increased ownership, trust, and accountability. This suggests that incorporating participatory mechanisms such as community consultations, local governance structures, and stakeholder engagement platforms can significantly enhance the effectiveness of clean energy initiatives. In the broader context of sustainable development, community inclusion also promotes social equity and empowerment, key components of Sustainable Development Theory. By giving communities a voice in energy planning, policies become more responsive and inclusive, thereby improving both adoption rates and developmental outcomes.

The findings on policy inconsistencies and weak institutional frameworks reveal a disconnect between policy formulation and implementation in Nigeria's energy sector. While policies such as the National Renewable Energy and Energy Efficiency Policy (NREEEP) demonstrate a commitment to clean energy, their limited impact reflects challenges related to governance, funding, and coordination.

This observation is consistent with the work of the International Energy Agency (IEA, 2022), which emphasizes that effective energy transitions require coherent and well-implemented policy frameworks. The absence of strong regulatory incentives and enforcement mechanisms in Nigeria has created an enabling environment that is insufficiently supportive of renewable energy investments.

From a theoretical perspective, this undermines the principles of Sustainable Development Theory, which advocates for integrated and well-coordinated institutional systems. It also weakens the diffusion process described by Rogers (2003), as policy environments play a crucial role in facilitating or constraining innovation adoption.

The implication is that Nigeria must move beyond policy formulation to focus on implementation, monitoring, and evaluation. Strengthening institutional capacity, ensuring policy coherence, and fostering inter-agency collaboration are essential for achieving meaningful progress in clean energy utilisation.

The study identifies high initial costs and infrastructural deficits as major barriers to clean energy adoption. This finding reflects broader structural challenges within Nigeria's economy and energy sector. Although renewable energy technologies offer long-term economic benefits, the upfront investment required poses a significant obstacle for low-income households.

This aligns with Aliyu et al. (2015), who highlight the financial constraints associated with renewable energy deployment in developing countries. The lack of accessible financing options, such as credit schemes and subsidies, further exacerbates the problem.

Infrastructural limitations, including inadequate grid systems and maintenance services, also hinder the scalability of clean energy solutions. These challenges underscore the need for increased investment in energy infrastructure and the development of innovative financing mechanisms, such as public-private partnerships (PPPs).

From a sustainability perspective, addressing these barriers is critical for ensuring equitable access to clean energy. Without deliberate efforts to make renewable technologies affordable and accessible, the benefits of clean energy utilisation will remain concentrated among a limited segment of the population.

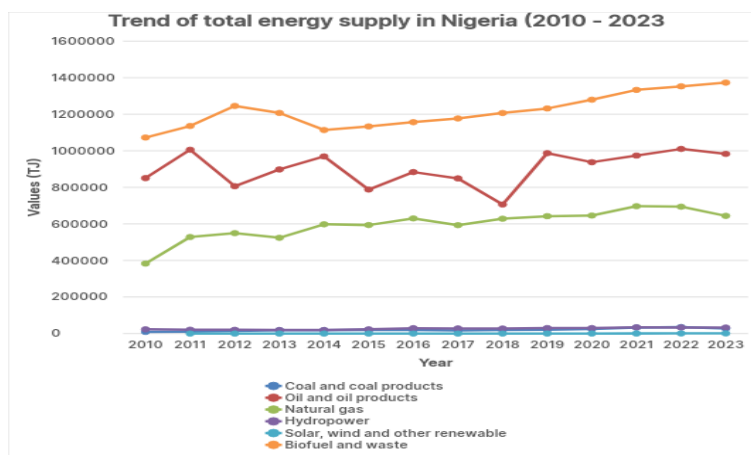


Figure 3: Total energy supplied in Nigeria from 2010 to 2023 (Source: Energy Access (EA), 2026)

The trend indicates that despite slight growth in renewable energy, fossil fuel dependence persists, reflecting a slow and limited transition toward clean energy in Nigeria.

The continued reliance on traditional biomass for energy needs highlights the persistent issue of energy poverty in Nigeria. This finding underscores the failure of existing energy systems to provide reliable and sustainable energy access, particularly in rural areas.

The environmental and health implications of this reliance are significant, including deforestation, indoor air pollution, and increased greenhouse gas emissions (UNDP, 2020). This situation contradicts the goals of climate change mitigation and sustainable development, as it perpetuates environmental degradation and social inequality.

Within the framework of Sustainable Development Theory, this finding emphasizes the need for a balanced approach that simultaneously addresses environmental protection and human development. Expanding access to clean energy in underserved communities is therefore not only an environmental imperative but also a socio-economic necessity.

One of the most significant findings of this study is the effectiveness of integrated approaches that combine community inclusion, mass education, and supportive public policies. This holistic strategy reflects the interplay between the three theoretical frameworks underpinning the study.

The success of such approaches demonstrates that clean energy utilisation is not merely a technical issue but a multidimensional challenge that requires coordinated action across social, educational, and institutional domains. By aligning these elements, it becomes possible to overcome existing barriers and accelerate the transition to sustainable energy systems.

This finding reinforces the argument by Sovacool (2014) that energy transitions are inherently socio-technical processes that require both technological innovation and social transformation. In the Nigerian context, this means that policies must be designed to simultaneously address awareness gaps, community engagement, and structural constraints.

5.3 Recommendations

Drawing from the findings and discussion of this study, the following recommendations are proposed to enhance clean energy utilisation for climate change mitigation and sustainable development in Nigeria. These recommendations emphasize the integration of community inclusion, mass education, and effective public policy as critical drivers of a successful energy transition.

i. **Strengthening Mass Education and Energy Literacy**

There is an urgent need to develop and implement comprehensive mass education programs aimed at improving energy literacy across Nigeria. Public awareness campaigns should be intensified through multiple channels, including radio, television, social media, and community outreach initiatives. These programs should focus on the benefits, accessibility, and usability of clean energy technologies, particularly targeting rural and underserved populations.

Furthermore, clean energy education should be integrated into formal educational curricula at primary, secondary, and tertiary levels. This will foster early awareness and build a generation that is knowledgeable about sustainable energy practices. Informal education mechanisms, such as workshops, seminars, and community demonstrations, should also be encouraged to reach a wider audience (Ockwell et al., 2009).

ii. **Promoting Community Inclusion and Participatory Energy Governance**

To enhance the sustainability and acceptance of clean energy initiatives, policymakers and development actors must prioritize community inclusion in energy

planning and implementation. This can be achieved by establishing participatory frameworks that actively involve local communities in decision-making processes. Community-based energy projects, such as cooperative solar mini-grids, should be promoted to foster local ownership and accountability. Traditional institutions, community leaders, and local organizations should be engaged as key stakeholders in energy programs to ensure that interventions are culturally appropriate and context-specific (Chambers, 1997).

In addition, capacity-building initiatives should be implemented to empower communities with the technical and managerial skills required to operate and maintain clean energy systems.

iii. Enhancing Policy Coherence and Institutional Effectiveness

The Nigerian government should prioritize the review and harmonization of existing energy policies to eliminate inconsistencies and improve implementation effectiveness. Policies such as the National Renewable Energy and Energy Efficiency Policy (NREEEP) should be strengthened with clear implementation strategies, measurable targets, and robust monitoring and evaluation mechanisms.

Institutional capacity should be enhanced through training, resource allocation, and inter-agency collaboration to ensure efficient policy execution. Regulatory frameworks should also be reinforced to provide a stable and predictable environment for clean energy investments (IEA, 2022).

Additionally, decentralization of energy governance should be considered to allow state and local governments to play a more active role in promoting clean energy initiatives.

iv. Expanding Access to Financing and Economic Incentives

To address the financial barriers associated with clean energy adoption, innovative financing mechanisms should be developed and expanded. These may include subsidies and tax incentives for renewable energy technologies, low-interest loans and microcredit schemes for households and small businesses, pay-as-you-go (PAYG) models to reduce upfront costs, public-private partnerships (PPPs) to mobilize investment in clean energy infrastructure and financial institutions should be encouraged to develop tailored products that support renewable energy investments, particularly for low-income and rural populations (World Bank, 2021).

v. Investing in Energy Infrastructure and Technological Innovation

Significant investment is required to improve energy infrastructure and support the deployment of clean energy technologies. This includes the development of decentralized energy systems such as mini-grids and off-grid solar solutions, which are particularly suitable for rural areas.

Research and development (R&D) should also be promoted to encourage innovation in renewable energy technologies that are affordable, efficient, and adapted to local conditions. Collaboration with academic institutions, research centers, and private sector actors can facilitate technological advancement and knowledge transfer.

vi. Addressing Energy Poverty through Targeted Interventions

Targeted interventions are needed to address energy poverty, especially in rural and marginalized communities. Government and development partners should prioritize the provision of clean and affordable energy solutions for cooking, lighting, and productive uses.

Programs promoting clean cookstoves and alternative fuels should be scaled up to reduce reliance on traditional biomass and mitigate associated health and environmental risks (UNDP, 2020). These interventions should be accompanied by awareness campaigns and financial support to ensure widespread adoption.

vii. Strengthening Public-Private Partnerships (PPPs)

Public-private partnerships should be leveraged as a strategic tool for accelerating clean energy deployment in Nigeria. The government should create an enabling environment that encourages private sector participation through transparent policies, risk-sharing mechanisms, and investment incentives.

Collaboration between government agencies, private companies, non-governmental organizations, and international development partners can enhance resource mobilization, technical expertise, and project implementation capacity.

viii. Establishing Monitoring, Evaluation, and Accountability Mechanisms

Effective monitoring and evaluation (M&E) systems should be established to track the progress of clean energy initiatives and ensure accountability. This includes the development of performance indicators, regular reporting, and impact assessments.

Data collection and management systems should be strengthened to provide reliable information for decision-making and policy adjustments. Transparency and accountability in the implementation of energy projects will enhance public trust and encourage stakeholder participation.

6. Conclusion

In conclusion, the study maintains that rethinking clean energy utilisation in Nigeria is both a necessity and an opportunity for advancing climate change mitigation and achieving sustainable development goals. This can be achieved by prioritizing inclusive governance, expanding educational outreach, and strengthening policy frameworks, Nigeria can unlock its renewable energy potential and transition towards a more resilient, equitable, and environmentally sustainable energy future.

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